SellerMargin

Detailed Product Design Specification

Know exactly how much you make on every sale.

Category	E-commerce / Seller Tools / Financial Utility
Primary platform	Responsive web application
Primary audience	Etsy, eBay, Shopify, Amazon Handmade, craft-fair, and independent online sellers
Business model	Freemium + one-time paid tools; optional Pro tier later
Architecture principle	Client-first, deterministic calculations, near-zero marginal infrastructure cost

1. Product Vision

SellerMargin is a fast, privacy-friendly profit and pricing calculator for small online sellers.

A seller enters selling price, product cost, packaging, shipping, marketplace, advertising, discounts, and other transaction costs. SellerMargin immediately answers the core question: How much money do I actually keep?

The product goes beyond a basic calculator by answering questions such as: What should I charge? Can I afford a 20% sale? How much does free shipping cost me? What price gives me a 40% margin? How many units must I sell to make $1,000 per month? Which products are actually profitable? Which marketplace produces better transaction economics for this item?

The initial product should require no AI, no account, no backend calculation engine, and almost no infrastructure. Most calculations happen entirely in the browser.

2. Problem

Small sellers often mistake sale price minus product cost for profit. Marketplace fees, payment processing, advertising, shipping, packaging, discounts, and fulfillment can materially reduce actual transaction profit.

SellerMargin makes those economics visible before the seller commits to a price, promotion, shipping strategy, or advertising decision.

3. Product Positioning

Do not position SellerMargin as accounting software, an AI-powered e-commerce financial platform, or a complete seller business-management suite.

Position it as: Know what you actually make.

Supporting copy: Calculate fees, shipping, advertising, materials and discounts - before you price your product.

The experience should feel closer to a high-quality calculator and decision tool than traditional accounting software.

4. Why This Product Is Commercially Attractive

SellerMargin has favorable economics: strong search intent, near-zero marginal cost for core calculations, no LLM requirement, no storage requirement for free users, no mandatory accounts, relatively low regulatory complexity, and a value proposition directly connected to seller income.

Potential acquisition queries include Etsy fee calculator, Etsy profit calculator, eBay fee calculator, Etsy pricing calculator, craft pricing calculator, profit margin calculator, and free shipping calculator.

5. Primary Users

- Handmade sellers producing jewelry, candles, art, clothing, home decor, personalized goods, or similar products.
- Resellers buying inventory and reselling on marketplaces such as eBay.
- Small Shopify merchants who need pricing and margin tools without full accounting software.
- Craft-fair sellers comparing in-person and online economics.
- New sellers searching for guidance such as how much should I charge on Etsy?

6. Core Jobs To Be Done

- Tell me how much I actually make from this sale.
- Calculate marketplace fees automatically.
- Tell me what price I need for my desired profit margin.
- Show what happens when I run a discount.
- Compare different marketplaces.
- Tell me whether free shipping makes financial sense.
- Analyze my entire product catalog.

7. Product Architecture Philosophy

SellerMargin should be client-first. The browser performs fee calculations, margin calculations, price simulations, marketplace comparisons, charts, and CSV parsing where practical.

The server should be reserved for payments, optional authentication, entitlement validation, and optional cloud persistence. This keeps operating costs extremely low.

8. Landing Page

Hero: Know what you actually make on every sale.

Subheadline: Calculate fees, shipping, advertising, materials and discounts - before you price your product.

Primary CTA: Calculate My Profit.

No signup should be required. The calculator should appear directly on the landing page with starter fields for selling price, marketplace, and product cost.

9. Instant Calculator

Inputs: selling price, marketplace, product cost, seller shipping cost, customer shipping charge, packaging, advertising, discount, and optional other costs.

Seller shipping cost and customer shipping charge must be separate fields.

After initial interaction, results update reactively without requiring a Calculate button.

10. Primary Result

Display revenue, marketplace fees, payment fees, product cost, packaging, shipping, transaction-level profit, and profit margin.

The core result must be visually prominent and immediately understandable.

11. Profit Definitions

Revenue: money received from the customer before costs.

Gross profit: revenue minus direct product cost.

Net contribution profit: revenue minus marketplace fees and seller-entered variable costs.

Avoid calling the result accounting or tax net income. Include a tooltip explaining that SellerMargin calculates transaction-level profit and does not replace accounting or tax records.

12. Visual Money Flow

Provide a waterfall-style visualization that starts with sale revenue and subtracts fees, materials, packaging, shipping, and other costs until the final profit is reached.

This should make the calculation intuitive even for users without financial training.

13. Marketplace Fee Engine

Represent marketplace fees through a versioned configuration model rather than scattered constants.

Each configuration should include marketplace, effective date, fee definitions, official source, and last-verified date.

Marketplace fee rules change. The UI should expose the last verification date and provide a way to inspect fee assumptions.

14. Etsy Calculator

Support relevant fee components such as listing fee, transaction fee, payment processing, shipping-related fee treatment, and advertising/offsite fees when applicable.

Exact formulas must be implemented from current official Etsy documentation at development time. Do not hard-code percentages from this product specification.

15. eBay Calculator

Allow product category, sale price, shipping charged, seller shipping cost, applicable final-value fee, and promoted-listing rate.

Exact formulas must come from current official eBay documentation.

16. Shopify Calculator

Support relevant plan/payment arrangement, payment processing, transaction fees where applicable, product cost, fulfillment, and advertising.

Do not model Shopify as having one universal fee structure.

17. Custom Marketplace

Allow users to enter a platform percentage, fixed transaction fee, and listing fee so SellerMargin remains useful for unsupported marketplaces.

18. Pricing Calculator

Reverse the normal calculation. Given product cost, packaging, shipping, marketplace, and a target margin, calculate the approximate minimum selling price required to achieve that margin.

Show the resulting estimated profit and margin at the suggested price.

19. Target Profit Mode

Allow the seller to enter a desired dollar profit per item and calculate the selling price required to achieve it under the selected marketplace and costs.

20. Discount Simulator

Compare normal-price economics against a user-selected discount such as 10%, 20%, or custom.

Show revenue, profit, margin, profit reduction, and the approximate increase in sales required to generate the same contribution profit as the undiscounted scenario.

21. Break-Even Discount

Calculate the maximum approximate discount before transaction-level profit reaches zero, based on the seller's entered costs and applicable fee rules.

22. Free Shipping Simulator

Compare customer-paid shipping with a free-shipping scenario, including cases where the seller raises product price to absorb shipping.

Explain that free shipping is a pricing strategy, not free fulfillment.

23. Advertising Simulator

Allow ad spend, attributed sales, and average order value. Calculate revenue, advertising spend, product/fulfillment costs, marketplace fees, ROAS, and - more importantly - profit after advertising.

24. Marketplace Advertising Scenario

Where marketplace advertising fees are conditional, compare organic-sale profit against ad-attributed-sale profit so acquisition cost becomes visible.

25. Marketplace Comparison

Allow one product to be compared across supported marketplaces. Show sale price, platform/payment fees, product cost, shipping, transaction profit, and margin.

Clearly state that this compares transaction economics only; it does not account for differences in customer acquisition, marketplace demand, or conversion.

26. Monthly Income Goal Calculator

Given profit per product and a monthly profit target, calculate approximate sales required per month, week, and day.

27. Hourly Earnings

For handmade products, optionally capture production time and calculate effective transaction profit per production hour.

28. Labor Cost Mode

Allow advanced users to specify a desired hourly labor rate and production time. Treat the resulting labor cost as part of product economics when calculating target prices and margins.

29. Batch Product Analyzer

This is the strongest initial paid feature.

Allow CSV upload containing fields such as SKU, price, product cost, packaging, and shipping. Process locally where practical.

Return a table containing revenue, fees, costs, profit, margin, and target-price information for every product.

30. Profitability Flags

Allow the seller to configure a target margin. Classify products relative to that target using labels such as Above target, Below target, and Negative profit.

Do not impose universal profitability thresholds.

31. Pricing Opportunities

Identify products below the seller's target margin and calculate the approximate price required to reach the target, while explicitly noting that price changes may affect demand.

32. What-If Mode

Create a global scenario engine. Examples: shipping increases 10%, marketplace fee increases 1%, store-wide 20% sale, material costs increase 15%, or prices increase 8%.

Immediately recalculate product or catalog economics. This should evolve SellerMargin from a calculator into a lightweight business simulator.

33. Scenario Comparison

Allow multiple named scenarios such as Current, Holiday Sale, Free Shipping, and Price Increase.

Compare price, profit, margin, and relevant costs side by side.

34. Product Saving

Free users should not require an account. Persist saved calculations locally through localStorage or IndexedDB.

Use messaging such as: Saved products stay in this browser unless you choose cloud sync.

35. Shareable Calculation

Provide a safe Copy Calculation feature that produces a human-readable summary.

If URL-encoded sharing is implemented, clearly warn users before embedding potentially sensitive business economics in a URL.

36. Paid Product: Seller Profit Pack

Free: unlimited individual calculations including profit calculator, marketplace fees, margin, pricing calculator, discount simulator, free-shipping simulator, and monthly sales goal.

Seller Profit Pack - recommended initial price: $7 one-time.

Paid features: batch CSV analysis, catalog profitability report, bulk target-price recommendations, what-if scenarios, downloadable CSV/Excel report, and printable PDF summary.

37. Why Keep The Calculator Free

The calculator is the acquisition channel. Users arriving from search should receive the answer immediately.

Do not require account creation to see basic results. Monetize workflow acceleration and batch analysis rather than basic arithmetic.

38. Potential Pro Tier Later

Only introduce recurring pricing after observing recurring usage.

Possible SellerMargin Pro: approximately $6/month or $49/year, with saved cloud catalog, sync, unlimited batch analysis, historical margin tracking, supplier cost changes, multiple stores, and monthly dashboards.

Do not build this for MVP.

39. No AI In MVP

Do not use an LLM merely to make the product appear AI-enabled.

The core problem is mathematical. Deterministic TypeScript is cheaper, faster, more accurate, explainable, and testable.

This is an intentional product and engineering decision.

40. Optional Lightweight AI Later

Potential AI use cases: mapping unusual CSV column names to SellerMargin fields, converting plain-English scenario requests into deterministic parameters, and explaining why one product is less profitable than another.

AI may interpret intent or explain structured results. It must not be responsible for financial arithmetic.

41. Fee Accuracy

Marketplace fee accuracy is a core trust dependency.

Every marketplace configuration should expose marketplace, fee name, rate/formula, effective date, official source, and last verified date.

Provide a View fee assumptions interaction.

42. Fee Rule Versioning

Preserve historical fee-rule versions instead of overwriting them. Example version identifiers may be marketplace-and-effective-period based.

This allows calculations to remain reproducible when marketplaces change fees.

43. Taxes

Distinguish marketplace-collected/remitted sales tax from seller income taxes.

Do not calculate income tax in MVP. State clearly that SellerMargin calculates transaction economics and does not calculate income taxes.

44. Currency

MVP should support USD only.

International expansion can later add CAD, GBP, EUR, AUD, and country-specific marketplace/payment/tax behavior.

45. Mobile UX

Mobile is first-class. Keep the initial form compact: selling price, marketplace, product cost, shipping, and prominent profit/margin results.

Place less common fields behind a More Costs control.

46. Desktop UX

Use a split layout with inputs on one side and live results, margin, money flow, and pricing recommendations on the other.

47. Interaction Quality

Do not require form submission. Use reactive calculation.

Every input change should update profit, margin, breakdown, visualization, and target-price results with effectively instant perceived latency.

48. URL Structure

Recommended functional routes: /, /etsy-profit-calculator, /ebay-profit-calculator, /shopify-profit-calculator, /pricing-calculator, /discount-calculator, /free-shipping-calculator, /craft-pricing-calculator, /profit-margin-calculator.

Each route should contain real interactive functionality rather than thin SEO content.

49. SEO Strategy

Target high-intent marketplace, pricing, margin, shipping, and promotion queries.

Every SEO landing page should provide immediate utility and naturally lead users into related SellerMargin tools.

50. Programmatic SEO - Use Carefully

Future category-specific pages such as how to price candles, handmade jewelry, crochet items, or soap may reuse the calculator with meaningful examples.

Do not generate hundreds of low-quality AI pages.

51. Privacy

Basic calculations should never need to leave the browser.

Prominently state: Your pricing data stays on your device.

Process batch CSV locally whenever technically practical to avoid storing SKUs, margins, costs, supplier economics, or other sensitive seller data.

52. Analytics

Track behavioral events such as calculator view, marketplace selection, advanced-cost usage, pricing-mode usage, discount simulation, shipping simulation, batch-tool view, batch upload start, checkout, and purchase.

Do not send selling price, product costs, profit, SKU, or catalog contents to general analytics.

53. Performance

Target an essentially instant calculator load, minimal JavaScript, no network request for calculations, and offline capability after initial load if PWA support is implemented.

54. Accessibility

Target WCAG AA.

Do not communicate financial status through color alone. Pair visual treatment with explicit labels such as Above target, Below target, and Negative profit.

55. PWA Potential

SellerMargin is a strong PWA candidate. Individual calculations should eventually work offline after installation.

This is useful at craft fairs, supplier stores, shipping counters, and inventory-planning sessions.

56. Export

Paid batch users should be able to export CSV with SKU, selling price, total fees, total costs, profit, margin, and target price.

Support formatted Excel and a printable PDF catalog profitability summary.

57. Testing Requirements

Maintain comprehensive unit tests because business logic is deterministic.

For each marketplace test zero price, normal sale, shipping, customer-paid shipping, free shipping, advertising, discounts, fixed fees, percentage fees, rounding, and boundary conditions.

Marketplace configurations should have regression fixtures.

58. Precision

Use integer minor currency units internally where practical, e.g. $35.00 represented as 3500 cents.

Avoid floating-point artifacts and apply marketplace-specific rounding rules where required.

59. Calculation Explainability

Every result should be inspectable. Clicking marketplace fees should reveal the individual components such as transaction fee, payment processing, and listing fee.

Do not present unexplained magic totals.

60. Error States

If costs exceed revenue, show a clear message such as You're losing $3.42 per sale.

If selling price is zero, do not calculate an invalid margin; ask the user to enter a selling price.

61. Technical Stack

- Frontend: React + TypeScript.
- Calculation engine: pure TypeScript module/package separated from UI.
- Hosting: inexpensive edge/static platform such as Cloudflare Workers/static assets.
- Payments: Stripe.
- Analytics: privacy-conscious.
- No database is required for MVP.

62. Calculation Engine Architecture

Recommended conceptual flow: TransactionInput -> MarketplaceFeeEngine -> CostEngine -> ProfitEngine -> MarginEngine -> ScenarioEngine -> CalculationResult.

The UI must not implement marketplace formulas directly.

63. Marketplace Adapter Interface

Define a clean marketplace-calculator abstraction that accepts normalized transaction input and returns a fee breakdown plus metadata.

Implement adapters for Etsy, eBay, Shopify, Direct Sale, and Custom as supported.

64. Scenario Engine

A scenario should transform base inputs and then call the same deterministic calculation engine.

Do not duplicate financial formulas for individual scenarios.

65. Zero Backend Mode

The entire free application should continue to calculate if the backend is unavailable.

Only paid entitlement, optional sync, and other server-dependent features should require backend interaction.

66. Security

Implement a strong Content Security Policy, dependency auditing, secure payment redirects, webhook signature verification, and ensure secrets never enter the client bundle.

CSV handling must protect against oversized/malformed files and spreadsheet formula injection in exported files.

67. Monetization Validation

Before building the paid batch system, launch free calculators with an Analyze your entire catalog - $7 CTA.

Measure clicks and checkout intent. If demand is weak, avoid spending weeks on batch workflows.

68. MVP Scope

- Marketplace support: Etsy, eBay, Direct/Custom.
- Calculators: profit, target price, target profit, discount, free shipping, monthly sales goal.
- UX: mobile, desktop, local persistence, transparent fee assumptions.

- Growth: marketplace-specific URLs, SEO metadata, structured data where appropriate.
- Monetization test: Seller Profit Pack CTA.

69. Explicit MVP Non-Goals

- Do not build marketplace OAuth integrations, bookkeeping, inventory management, order management, shipping labels, tax filing, invoices, supplier management, CRM, AI chatbot, native applications, team collaboration, or subscription billing.

70. Development Milestones

- Milestone 1 - Core Math Engine: revenue, costs, profit, margin, and exhaustive tests.
- Milestone 2 - Etsy Adapter: implement current official fee model.
- Milestone 3 - eBay Adapter: implement using the same architecture.
- Milestone 4 - Calculator UI: responsive reactive interface.
- Milestone 5 - Reverse Pricing: target margin and target profit.
- Milestone 6 - Simulators: discount and free shipping.
- Milestone 7 - SEO Pages: platform-specific calculators.
- Milestone 8 - Monetization Validation: add Profit Pack CTA and launch.
- Milestone 9 - Batch Tool: build only after evidence of demand.

71. MVP Definition of Done

SellerMargin is ready when a seller can open the site on a phone, enter price, marketplace, product cost, shipping, packaging, and advertising, and immediately understand revenue, marketplace fees, total costs, transaction profit, and margin.

They can then test a discount or target margin and receive an immediate deterministic answer without creating an account, uploading data, waiting for AI, or paying.

72. Business Experiment

Build and deploy the free calculator first. Add Analyze Your Entire Catalog - $7 as the paid CTA.

Measure the funnel from calculator visitors to Profit Pack clicks, checkout starts, and purchases. This provides real willingness-to-pay evidence before substantial paid-feature development.

73. Potential Revenue Paths

Transactional: $7 Seller Profit Pack - preferred initial model.

Subscription: approximately $49/year SellerMargin Pro - only after recurring usage is demonstrated.

Affiliate: potentially shipping, packaging, e-commerce, or business-service partners, provided affiliate relationships never influence calculator outputs.

74. Portfolio Story

SellerMargin demonstrates domain modeling, financial precision, marketplace adapter architecture, complex reactive state, scenario modeling, visualization, performance, client-side privacy, regression testing, and product judgment.

A key engineering story is intentionally not using an LLM for deterministic financial calculations.

75. What Makes SellerMargin Defensible

The long-term differentiation is not merely a profit calculator; it is a what-if engine for small sellers.

A seller defines a product once and can immediately explore discounts, shipping increases, material-cost changes, price changes, ad-attributed sales, marketplace changes, and target-margin scenarios.

76. Recommended Product Identity

Working name: SellerMargin.

Other names worth testing include TrueMargin, KeepRate, ActualProfit, MarginKit, and SellWise.

For an SEO-first launch, SellerMargin is a strong working name because the purpose is immediately understandable.

77. Final Product Test

The product succeeds when a seller enters a real product and discovers something actionable they did not realize - for example, that a 20% holiday discount reduces profit far more than expected, or that a target margin requires a materially different price.

The key user reaction should be: I didn't realize that.

78. Recommended Build Priority

Among the initial product concepts, SellerMargin has the lowest cost and fastest path to launch.

It is a strong candidate for testing the complete idea -> build -> SEO -> traffic -> first-dollar loop before investing heavily in document-heavy AI SaaS products.

Appendix A - Core Calculation Example

Line item	Example
Sale price	$35.00
Marketplace/payment fees	-$4.32
Product cost	-$9.00
Packaging	-$1.50
Shipping	-$6.00
Transaction profit	$14.18
Profit margin	40.5%

Appendix B - Guiding Engineering Rule

AI interprets; deterministic code calculates.

If lightweight AI is introduced later, it may map inputs, interpret natural-language scenarios, or explain structured results. Marketplace fees, profit, margin, pricing, and scenario arithmetic remain deterministic and testable.